



PROJECT-BASED LEARNING

PHASE-I EVALUATION

Smart Public Complaint Analyzer

Team ID: DSCPP-III-2026-T031

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026-27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T031

Semester: 3rd

Section: ML-1,ML-2,DS-1,C

Domain: AIML,AIDS,CORE

Team Members

1. *Devraj Singh Rawat – 2510370241*
2. *Stuti Mahajan – 2510010881*
3. *Archi Panwar – 2510012717*
4. *Rajat – 2510370572*

Mentor

Dr. Prabhdeep Singh

Interactions completed: _____

Problem Statement & Motivation

Problem Statement

Citizens report issues like garbage, potholes, broken street lights, and water leakage, but handling large numbers of complaints manually can cause delays, duplication, and incorrect routing. The project addresses the gap by providing a single structured system that categorizes, prioritizes, searches, routes, and tracks complaints efficiently using C data structures and C++ OOP concepts.

Motivation

- ▶ *Delayed complaint processing*
- ▶ *Inefficient resource utilization*
- ▶ *Faster complaint resolution*

Target Users / Stakeholders

- ▶ *Citizens, Government officials*
- ▶ *Municipal departments, System administrators*
- ▶ *Expected benefit*

Objectives

Project Objectives

1. *To provide easy complaint registration and status tracking for citizens.*
2. *To efficiently process, prioritize, sort, search, and route complaints using C.*
3. *including Data Structures in C and Object-Oriented Programming in C++..*
4. *To reduce duplicate complaints and manual workload through systematic processing.*
5. *To improve transparency and monitoring through an admin dashboard, reports, and feedback.*

Key Objective: *To efficiently prioritize, route, track, and manage public complaints from registration to resolution*

Proposed Solution

Proposed Approach

1. *Collect citizen complaints and issue details.*
2. *Categorize, detect duplicates, and calculate priority.*
3. *Assign complaints to relevant departments.*
4. *Store complaints, users, and status records.*
5. *Display complaint status and resolution updates.*

Key Features

- ▶ *Complaint Registration*
- ▶ *Automatic Categorization*
- ▶ *Duplicate Detection*
- ▶ *Priority Assignment*
- ▶ *Status Tracking*

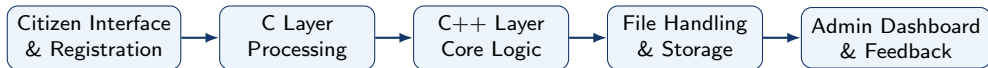
Clearly explain how the proposed solution addresses the identified problem.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	<i>Arrays, Linked Lists, Hashing, Queues, Searching, Sorting</i>	<i>Complaint Storage , Duplicate Detection , Priority Queue</i>
OOPs with C++	<i>Classes, Encapsulation, Inheritance, Polymorphism, STL</i>	<i>ComplaintProcessing, StatusTracking, DepartmentAssign-ment</i>

Requirement: Demonstrate meaningful integration of both subjects in **one** project.

System Architecture / Workflow



Major Components

- ▶ *C Engine Core*
- ▶ *System Controller Central Logic*
- ▶ *Persistence UI Layer*

Data / Control Flow

- ▶ *Citizen registers complaint; C layer stores, processes, searches and routes it.*
- ▶ *C++ layer manages entities using OOP, STL and exception handling.*
- ▶ *Admin assigns complaints to departments; reports and feedback update the system.*

Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C / C++*
- ▶ Database: *File Handling*
- ▶ Tools / Framework: *STL*
- ▶ IDE: *VS Code / Code::Blocks*
- ▶ Version Control: *Git / GitHub*

Development Methodology

1. **Requirement Analysis:** Identify complaint registration, categorization, search, routing and tracking requirements.
2. **System Design:** Design C/C++ modules, data structures, workflow and file storage.
3. **Implementation:** Develop C processing modules and C++ OOP, STL and exception handling.
4. **Testing:** Test registration, search, sorting, routing, file handling and error cases.
5. **Evaluation:** Evaluate accuracy, response time, reliability and overall effectiveness.

C handles processing, sorting, searching and routing, while C++ manages OOP-based entities, STL and exception handling.

Initial Progress & Team Contribution

Progress Achieved

- ▶ *Background study completed; key issues identified.*
- ▶ *System requirements for registration, categorization, priority and tracking identified.*
- ▶ *C/C++ architecture and complete system workflow prepared.*
- ▶ *Initial complaint processing, searching and routing module implemented.*
- ▶ *Data structures and file storage prepared.*

Individual Contribution

Member	Role	Contribution
Devraj Singh Rawat	Lead	Planning and coordination
Stuti Mahajan	Developer	C/C++ module development
Archi Panwar	DB/Testing	Database and testing
Rajat	Developer/Docs	Development and documentation

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Problem Analysis & Design
2. Phase-II: System Development
3. Phase-III: Testing & Implementation

Expected Outcomes

- ▶ Working complaint analysis system
- ▶ Automatic complaint categorization
- ▶ Priority and duplicate detection
- ▶ Department assignment and tracking

References

1. Complaint Management Research Papers
2. Data Structures & Algorithms Books
3. C/C++ and STL Documentation
4. Smart Civic Complaint Management System — research paper

Thank You
Questions & Suggestions